

RaceDay API Endpoint Plan

This document explains well enough the endpoints that are planned for the RaceDay application. The endpoints are authentication, user profiles, event management, event categories, participant enrolments, and race results. The access endpoints are controlled according to the user's role.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account and assigns the selected role.	Public	FirstName, LastName, Email, Password, PhoneNumber, RoleID	201 Created – User successfully registered. 400 Bad Request – Invalid or duplicate details.
POST	/api/auth/login	Authenticates a registered user using their email and password.	Public	Email, Password	200 OK – Authentication successful and token returned. 401 Unauthorized – Invalid credentials.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Returns the authenticated user's own profile information.	Organiser / Participant	None	200 OK – Profile returned. 401 Unauthorized – User is not authenticated.
PUT	/api/users/profile	Allows an authenticated user to update their own profile information.	Organiser / Participant	FirstName, LastName, PhoneNumber	200 OK – Profile updated. 400 Bad Request – Invalid information. 401 Unauthorized – Not authenticated.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns a list of available RaceDay events.	Organiser / Participant	None	200 OK – List of events.
GET	/api/events/{id}	Returns the details of a specific event.	Organiser / Participant	None	200 OK – Event returned. 404 Not Found – Event does not exist. 201 Created – Event created. 400 Bad Request – Invalid event data. 403 Forbidden – Participant attempted access.
POST	/api/events	Allows an organiser to create a new race event.	Organiser	EventName, Description, EventDate, Location, EventType	200 OK – Event updated successfully. 404 Not Found – Event does not exist. 403 Forbidden – User is not authorised.
PUT	/api/events/{id}	Updates an existing event managed by the organiser.	Organiser	EventName, Description, EventDate, Location, EventType	204 No Content – Event deleted successfully. 404 Not Found – Event does not exist. 403 Forbidden – User is not authorised.
DELETE	/api/events/{id}	Deletes an event managed by the organiser.	Organiser	None	404 Not Found – Event does not exist. 403 Forbidden – User is not authorised.

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories available for / a specific event.	Organiser Participant	None	200 OK – Categories returned. 404 Not Found – Event does not exist. 201 Created – Category created successfully.
POST	/api/events/{eventId}/categories	Creates a new category for an event.	Organiser	CategoryName	400 Bad Request – Invalid category data. 403 Forbidden – User is not authorised. 200 OK – Category updated successfully.
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	CategoryName	404 Not Found – Category does not exist. 204 No Content – Category deleted successfully.
DELETE	/api/categories/{id}	Deletes an existing event category.	Organiser	None	404 Not Found – Category does not exist.

5. Registrations

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/registrations	Allows a participant to enrol in an event by selecting a category.	Participant	EventID, CategoryID	201 Created – Registration successful. 400 Bad Request – Invalid registration. 409 Conflict – Participant already registered.
GET	/api/registrations/my	Retrieves the authenticated participant's registrations.	Participant	None	200 OK – Registrations returned.
GET	/api/events/{eventId}/registrations	Allows an organiser to view participants enrolled in an event they manage.	Organiser	None	200 OK – Registrations returned. 404 Not Found – Event does not exist. 403 Forbidden – User is not authorised.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Records a participant's finish time	Organiser	RegistrationID, FinishTime, Position	201 Created – Result recorded. 400 Bad

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		and finishing position.			Request – Invalid result. 404 Not Found – Registration does not exist.
GET	/api/results/my	Retrieves the authenticated participant's race results.	Participant	None	200 OK – Results returned.